

Chinese Remainder Theorem

1 Main Problem

Goal

The Chinese Remainder Theorem helps solve several remainder conditions at the same time:

$$\begin{aligned}x &\equiv a_1 \pmod{n_1}, \\x &\equiv a_2 \pmod{n_2}, \\&\vdots \\x &\equiv a_k \pmod{n_k}.\end{aligned}$$

The moduli must be pairwise relatively prime:

$$\gcd(n_i, n_j) = 1 \quad \text{whenever } i \neq j.$$

Main Idea

If the moduli are pairwise relatively prime, then the system has exactly one answer modulo

$$N = n_1 n_2 \cdots n_k.$$

If x_0 is the smallest nonnegative answer, then all integer answers are

$$x = x_0 + tN, \quad t \in \mathbb{Z}.$$

2 Theorem

Chinese Remainder Theorem

Suppose $n_1, n_2, \dots, n_k$ are pairwise relatively prime.

Then the system

$$x \equiv a_i \pmod{n_i} \quad (i = 1, 2, \dots, k)$$

has a unique solution modulo

$$N = n_1 n_2 \cdots n_k.$$

3 Construction Method

How to Build the Answer

For each congruence $x \equiv a_i \pmod{n_i}$:

$$M_i = \frac{N}{n_i}.$$

Then find y_i such that

$$M_i y_i \equiv 1 \pmod{n_i}.$$

Finally compute

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k \pmod{N}.$$

The number y_i is the inverse of M_i modulo n_i . This inverse exists because M_i and n_i are relatively prime.

CRT Checklist

1. Multiply the moduli: $N = n_1 n_2 \dots n_k$.
2. For each equation, compute $M_i = N / n_i$.
3. Find y_i so that $M_i y_i \equiv 1 \pmod{n_i}$.
4. Add all terms $a_i M_i y_i$.
5. Reduce the final sum modulo N .

4 Why the Construction Works

The term

$$a_i M_i y_i$$

is designed to affect only the i -th congruence.

Key Switch

For the i -th modulus:

$$M_i y_i \equiv 1 \pmod{n_i}.$$

For any other modulus n_j , where $j \neq i$:

$$M_i y_i \equiv 0 \pmod{n_j}.$$

Why does the second line happen? Since

$$M_i = \frac{N}{n_i},$$

the number M_i contains every modulus except n_i . So if $j \neq i$, then $n_j \mid M_i$, and therefore

$$M_i y_i \equiv 0 \pmod{n_j}.$$

5 Proof

Part 1: A solution exists

Define

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k \pmod{N}.$$

We must show that this x satisfies every congruence.

Look at one fixed modulus, say n_i . When we reduce x modulo n_i :

$$x \equiv a_1 M_1 y_1 + \cdots + a_i M_i y_i + \cdots + a_k M_k y_k \pmod{n_i}.$$

Every term except the i -th term becomes 0, because those M_j 's contain n_i as a factor:

$$a_j M_j y_j \equiv 0 \pmod{n_i} \quad (j \neq i).$$

So only the i -th term remains:

$$x \equiv a_i M_i y_i \pmod{n_i}.$$

But y_i was chosen so that

$$M_i y_i \equiv 1 \pmod{n_i}.$$

Therefore

$$x \equiv a_i \cdot 1 \pmod{n_i}.$$

So

$$\boxed{x \equiv a_i \pmod{n_i}}.$$

Since this works for every i , the constructed x is a solution.

Part 2: The solution is unique modulo N

Suppose x and z are both solutions of the same system.

Then for every i ,

$$x \equiv a_i \pmod{n_i} \quad \text{and} \quad z \equiv a_i \pmod{n_i}.$$

So

$$x \equiv z \pmod{n_i}.$$

That means

$$n_i \mid (x - z)$$

for every i .

Because the moduli $n_1, n_2, \dots, n_k$ are pairwise relatively prime, if each n_i divides $x - z$, then their product divides $x - z$:

$$N \mid (x - z).$$

Therefore

$$\boxed{x \equiv z \pmod{N}}.$$

So there is only one solution modulo N .

6 Worked Example

Problem

Solve

$$x \equiv 2 \pmod{5}$$

and

$$x \equiv 3 \pmod{13}.$$

The moduli 5 and 13 are relatively prime, so CRT applies.

$$N = 5 \cdot 13 = 65.$$

Step 1: Compute M_1 and M_2

$$M_1 = \frac{65}{5} = 13, \quad M_2 = \frac{65}{13} = 5.$$

Step 2: Find inverses

For the first congruence, find y_1 :

$$13y_1 \equiv 1 \pmod{5}.$$

Since

$$13 \equiv 3 \pmod{5}$$

and

$$3 \cdot 2 = 6 \equiv 1 \pmod{5},$$

we get

$$y_1 = 2.$$

For the second congruence, find y_2 :

$$5y_2 \equiv 1 \pmod{13}.$$

Since

$$5 \cdot 8 = 40 \equiv 1 \pmod{13},$$

we get

$$y_2 = 8.$$

Step 3: Combine

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{65}.$$

Here $a_1 = 2$ and $a_2 = 3$, so

$$x \equiv 2(13)(2) + 3(5)(8) \pmod{65}.$$

$$x \equiv 52 + 120 \pmod{65}.$$

$$x \equiv 172 \pmod{65}.$$

$$x \equiv 42 \pmod{65}.$$

Check

$$42 \equiv 2 \pmod{5}, \quad 42 \equiv 3 \pmod{13}.$$

7 Classic Three-Modulus Example

Problem

Solve

$$x \equiv 2 \pmod{3},$$

$$x \equiv 3 \pmod{5},$$

$$x \equiv 2 \pmod{7}.$$

The moduli 3, 5, 7 are pairwise relatively prime.

$$N = 3 \cdot 5 \cdot 7 = 105.$$

Congruence	n_i	a_i	$M_i = N/n_i$
1	3	2	35
2	5	3	21
3	7	2	15

Step 1: Find inverses

$$35y_1 \equiv 1 \pmod{3}.$$

Since $35 \equiv 2 \pmod{3}$ and $2 \cdot 2 \equiv 1 \pmod{3}$,

$$y_1 = 2.$$

$$21y_2 \equiv 1 \pmod{5}.$$

Since $21 \equiv 1 \pmod{5}$,

$$y_2 = 1.$$

$$15y_3 \equiv 1 \pmod{7}.$$

Since $15 \equiv 1 \pmod{7}$,

$$y_3 = 1.$$

Step 2: Combine

$$x \equiv 2(35)(2) + 3(21)(1) + 2(15)(1) \pmod{105}.$$

$$x \equiv 140 + 63 + 30 \pmod{105}.$$

$$x \equiv 233 \pmod{105}.$$

$$x \equiv 23 \pmod{105}.$$

All integer solutions are

$$x = 23 + 105t, \quad t \in \mathbb{Z}.$$

Check

$$23 \equiv 2 \pmod{3}, \quad 23 \equiv 3 \pmod{5}, \quad 23 \equiv 2 \pmod{7}.$$

8 Another Example

Problem

Solve

$$x \equiv 4 \pmod{5}$$

and

$$x \equiv 5 \pmod{11}.$$

$$N = 5 \cdot 11 = 55.$$

$$M_1 = \frac{55}{5} = 11, \quad M_2 = \frac{55}{11} = 5.$$

Find inverses:

$$11y_1 \equiv 1 \pmod{5}.$$

Since $11 \equiv 1 \pmod{5}$, we take

$$y_1 = 1.$$

$$5y_2 \equiv 1 \pmod{11}.$$

Since $5 \cdot 9 = 45 \equiv 1 \pmod{11}$, we take

$$y_2 = 9.$$

Now combine:

$$x \equiv 4(11)(1) + 5(5)(9) \pmod{55}.$$

$$x \equiv 44 + 225 \pmod{55}.$$

$$x \equiv 269 \pmod{55}.$$

$$x \equiv 49 \pmod{55}.$$

Check:

$$49 \equiv 4 \pmod{5}, \quad 49 \equiv 5 \pmod{11}.$$

9 When CRT Can Fail

Pairwise Coprime Condition

The simple CRT theorem needs pairwise relatively prime moduli.
If the moduli share a factor, the system may have no solution.

For example,

$$x \equiv 1 \pmod{4}, \quad x \equiv 2 \pmod{6}.$$

The first congruence says x is odd. The second congruence says x is even. So the system has no solution.

10 Final Review

Core Steps

$$N = n_1 n_2 \cdots n_k.$$

$$M_i = N/n_i.$$

$$\text{Choose } y_i \text{ so that } M_i y_i \equiv 1 \pmod{n_i}.$$

$$x \equiv a_1 M_1 y_1 + \cdots + a_k M_k y_k \pmod{N}.$$